

ECON 204 (Contemporary Economic Issues): In-Class Empirical Group Activity

Inflation, Cost of Living, and Monetary Policy in Canada

Format: groups of 3–4 **Time:** 45–60 minutes **Tool:** Excel or Google Sheets

Learning goals. In this activity you will (i) construct Canadian inflation from CPI data, (ii) connect inflation to cost-of-living changes, and (iii) evaluate how the Bank of Canada’s policy rate moves relative to inflation.

Data you will use (provided by the instructor). You will receive four files:

- `canada_cpi_allitems_monthly_2015_2025.csv`: all-items CPI level for Canada (monthly).
- `canada_policy_rate_target_monthly_eom_2015_2025.csv`: policy interest rate (“target rate”), end-of-month (monthly).
- `canada_avg_hourly_earnings_hourly_paid_monthly_2015_2025.csv`: a simple wage proxy (monthly).
- `canada_inflation_policy_wage_monthly_2015_2025.csv`: a cleaned file that merges the three series by month.

Part A. Organize the dataset (10 minutes)

Open the Excel template provided by the instructor. Confirm that the **Data** sheet contains these columns: `date`, `cpi`, `policy_rate`, `avg_hourly_earnings`.

Checkpoint A1. What is the first month in the dataset? What is the last month?

Part B. Construct inflation (10–15 minutes)

Use the CPI series to compute year-over-year inflation:

$$\pi_t = \left(\frac{CPI_t}{CPI_{t-12}} - 1 \right) \times 100.$$

In the template, this is computed on the **Calculations** sheet.

Checkpoint B1. Identify the month with the highest year-over-year inflation in 2015–2025. Record the month and the inflation rate.

Part C. Make two charts (10–15 minutes)

Create the following two charts (titles and axis labels matter):

- **Chart 1:** CPI level and inflation (YoY). If you use a secondary axis, explain what each axis measures.
- **Chart 2:** Inflation (YoY) and the policy rate. Use the same monthly date axis.

Checkpoint C1. In Chart 2, does the policy rate appear to rise *before* inflation peaks, *after* inflation peaks, or roughly *at the same time*? Give one sentence.

Part D. Cost-of-living interpretation (10 minutes)

Inflation is an economy-wide average, but cost-of-living pressure can be felt very differently across households. Use your charts to anchor your answers.

D1. Canadian cost-of-living channels. Pick **two** Canadian-relevant channels and write one paragraph on each, connecting them to your inflation surge period:

- food and groceries,
- housing costs (rent, mortgage interest),
- transportation and gasoline,
- services inflation,
- exchange rate and imported goods.

Channel 1 paragraph:

Channel 2 paragraph:

Part E. Optional extension (5 minutes)

Compute an approximate **real policy rate**:

$$r_t \approx i_t - \pi_t,$$

where i_t is the policy rate and π_t is year-over-year inflation.

E1. Identify one period where the approximate real policy rate is negative. In Canada, who is more likely to benefit from negative real rates (borrowers or savers), and why?

2-minute group share-out (if time permits)

Each group shares: (i) the month of peak inflation, (ii) whether policy appears to lag or lead, and (iii) one Canadian cost-of-living channel that is most salient in your charts.